

Lending Interaction Wings to Recommender Systems with Conversational Agents

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Content

- Problem Background
 - Conversational Recommender Systems
 - Current Challenges
- Architecture
 - Overall Framework
 - Online-Checking Module
 - Offline-Training Module
- Experiment
- Conclusion

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Conversational Recommender Systems

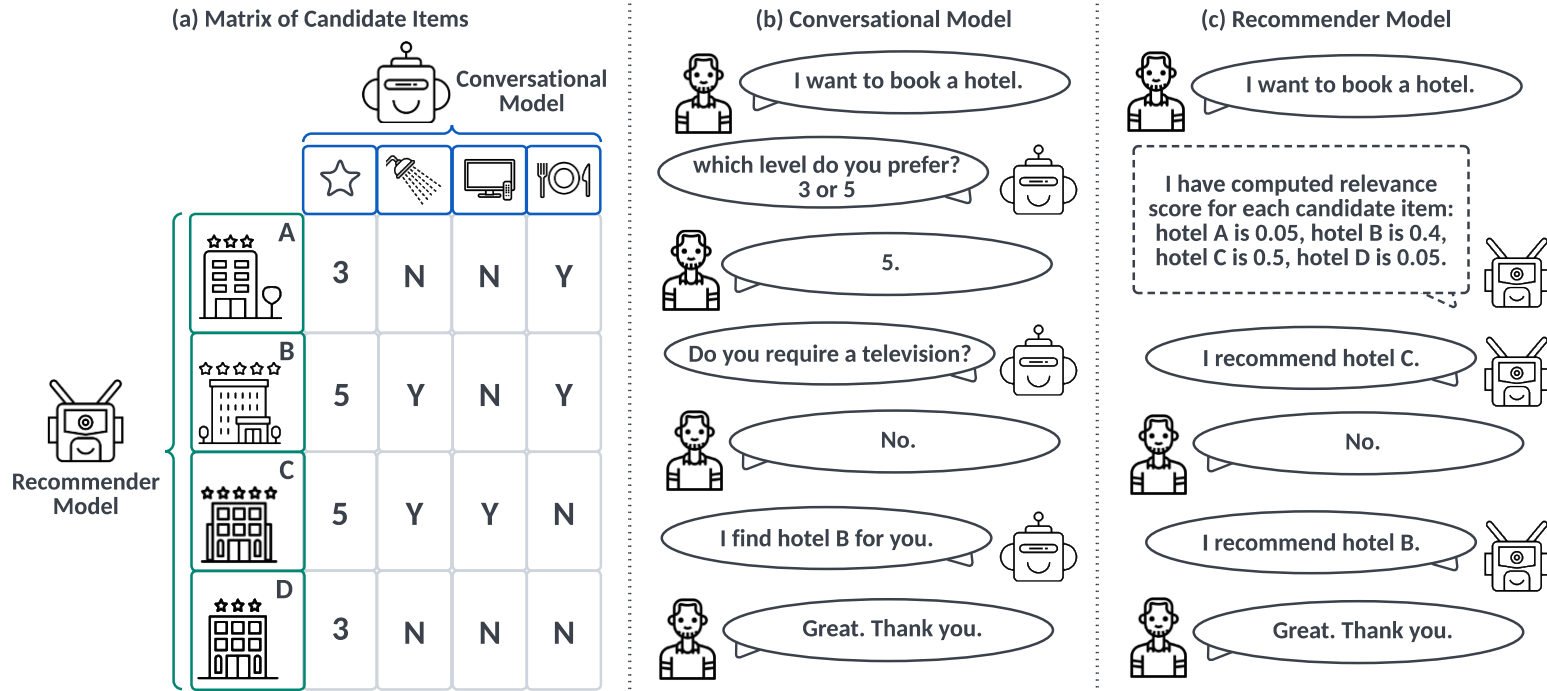


Figure: Comparisons between conversational model and (classical) recommender model.

Most currently deployed recommendation platforms are based on classical recommender models (assigning estimated score to each item according to historical data of users).

Conversational Recommender Systems

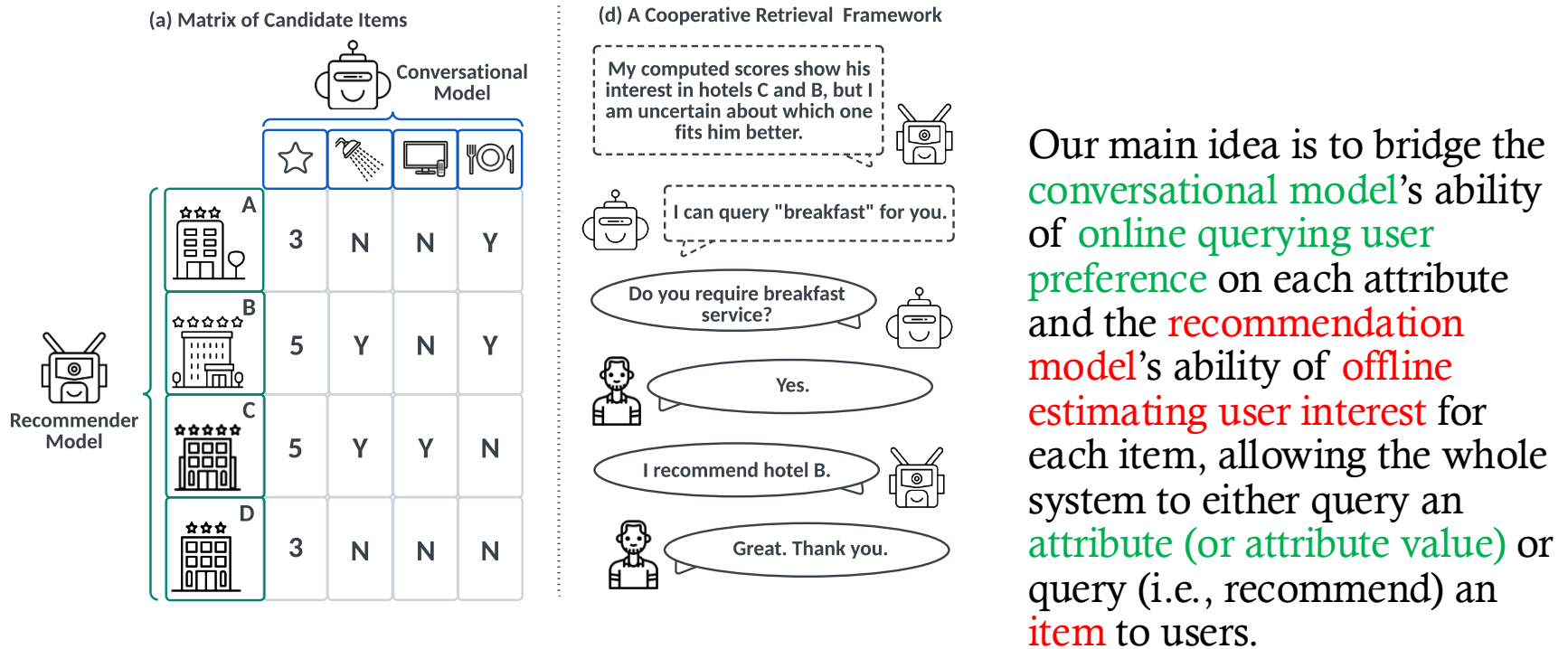


Figure: Cooperation between conversational model and (classical) recommender model.

Current Challenges

- [How to determine which one attribute (or attribute value) or one item to query]
- The key technique is how to jointly measure the reward (information gain) of querying **attribute values** and **items**.

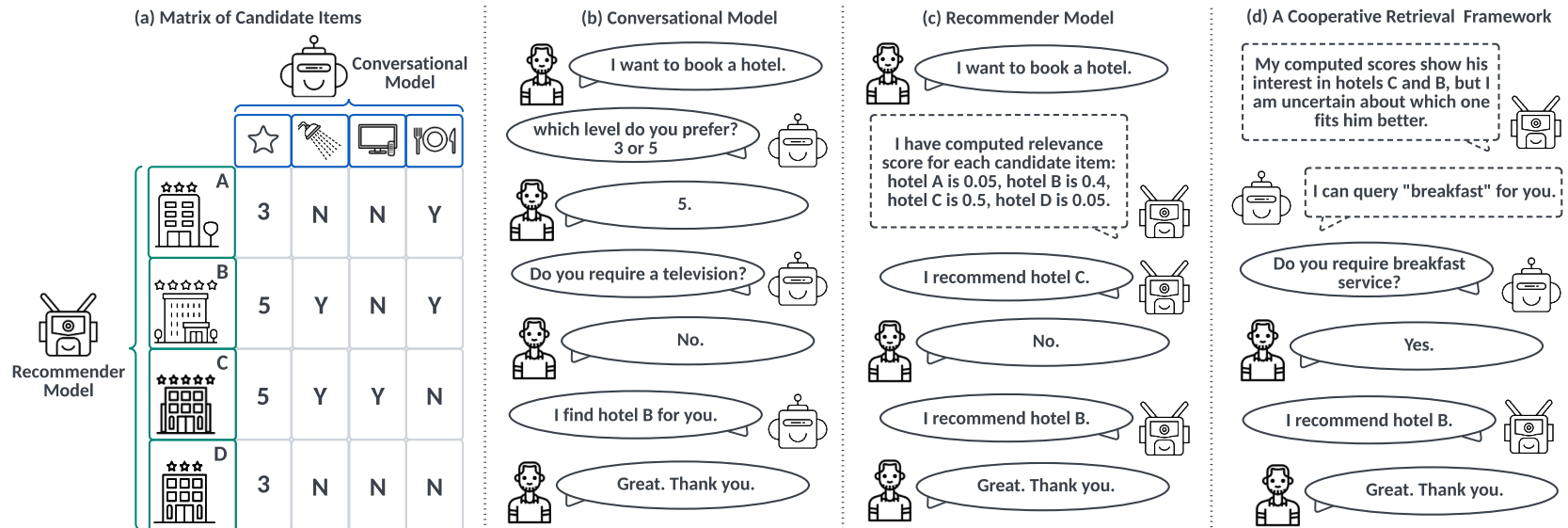


Figure: Cooperation between conversational model and (classical) recommender model.

Current Challenges

- [How to design the interaction mechanism with users]
- Previous literature such as [1][2] introduce **conversational module** together with **recommender** module, and jointly train them via reinforcement learning.

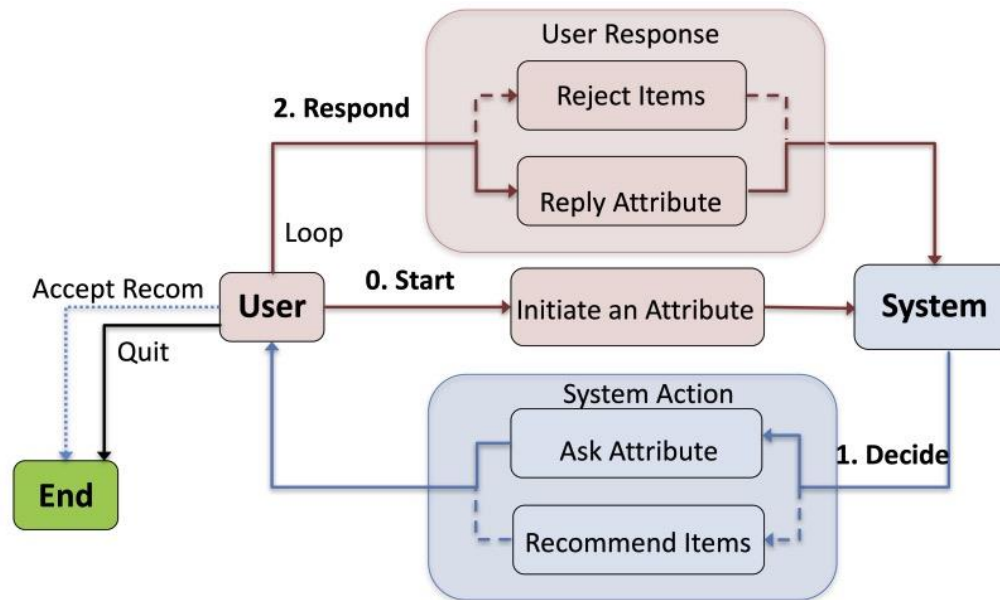


Figure: The workflow of multi-round conversational recommendation scenario.

Ref: Wenqiang Lei, et al. Estimation–Action–Reflection: Towards Deep Interaction Between Conversational and Recommender Systems. WSDM, 2020.

Ref: Yueming Sun, et al. Conversational Recommender System. SIGIR, 2018.

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Overall Framework

Bridge recommender systems and conversational agents through an offline-training and online-checking framework, only requiring **estimated scores** from recommender systems (offline-training) and **APIs** from conversational agent (online checking).

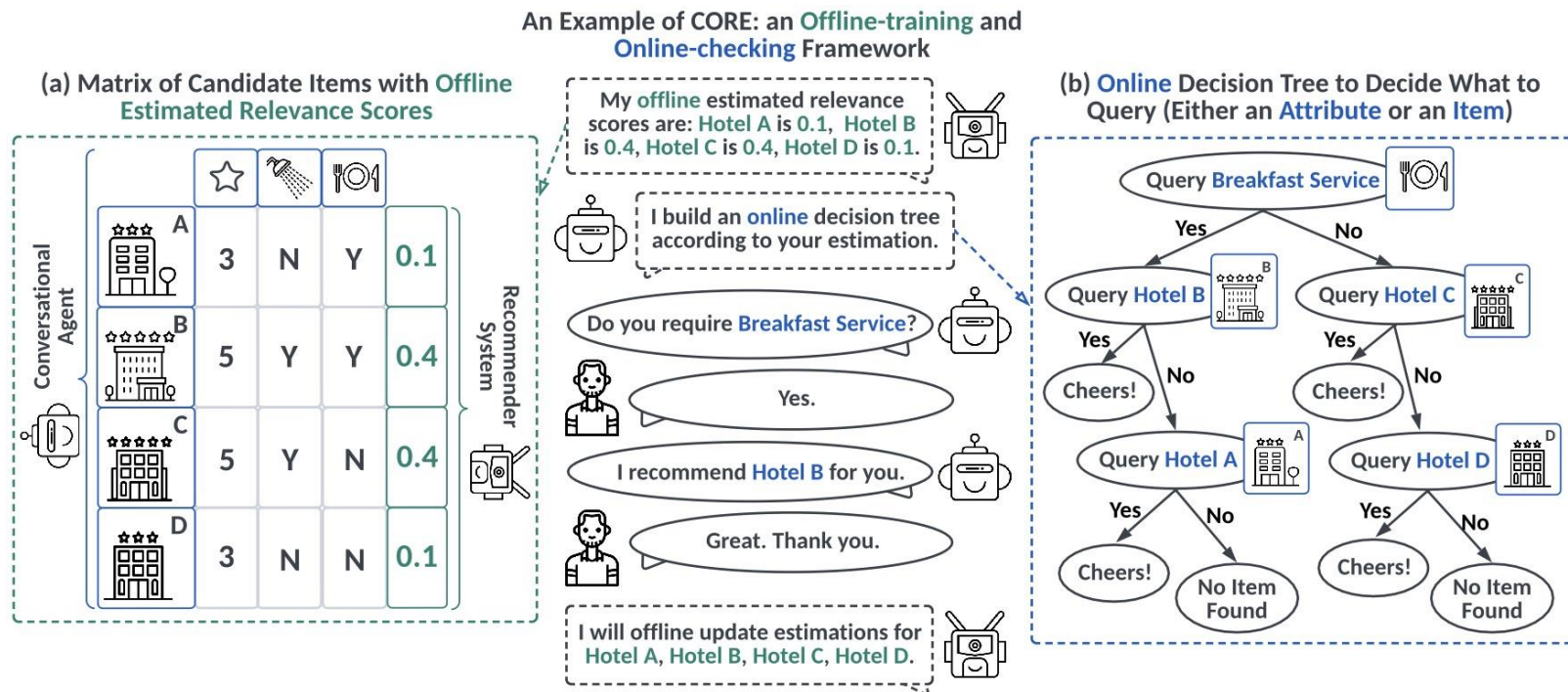


Figure: Overview of our offline-training and online-checking framework.

Online Decision Tree

- Given estimated **relevance scores** from an **offline-trained recommender system**, our certainty gain is defined as **how many relevance scores have been checked by querying an item or an attribute value**.

Definition 1 (Uncertainty and Certainty Gain). For the k -th turn, we define uncertainty, denoted as U_k , to measure how many estimated relevance scores are still unchecked, i.e.,

$$U_k := \text{SUM}(\{\Psi_{\text{RE}}(v_m) | v_m \in \mathcal{V}_k\}), \quad (1)$$

where $\Psi_{\text{RE}}(v_m)$ ¹ outputs the estimated relevance score for item v_m , $\mathcal{V}_k$ is the set of all the unchecked items after k interactions, and $\mathcal{V}_k$ is initialized as $\mathcal{V}_0 = \mathcal{V}$. Then, the certainty gain of k -th interaction is defined as $\Delta U_k := U_{k-1} - U_k$, i.e., how many relevance scores are checked at the k -th turn. Since our goal is to find **a target item** if $\Psi_{\text{CD}}(\cdot)$ successfully finds one at the k -th turn, then we set all the items checked, namely $U_k = 0$.

In this regard, our conversational agent is to minimize U_k at each k -th turn via removing those checked items from $\mathcal{V}_k$. Considering that online updating $\Psi_{\text{RE}}(\cdot)$ is infeasible in practice due to the high latency and computation costs, the objective of $\Psi_{\text{CD}}(\cdot)$ can be expressed as:

$$\min_{\Psi_{\text{RE}}^*} K, \text{ s.t., } U_K = 0, \quad (2)$$

where K is the number of turns, and Ψ_{RE}^* means that the recommender system is frozen. To this end, the design of our conversational agent could be organized as an uncertainty minimization problem.

Online Checking Module

- The resulting online decision tree algorithm is to greedily choose an attribute (value) or an item that can maximize the certainty gain:

$$a_{\text{query}} = \arg \max_{a \in \mathcal{V}_{k-1} \cup \mathcal{X}_{k-1}} \Psi_{\text{CG}}(\text{query}(a)),$$

(a) Matrix of Candidate Items with Offline Estimated Relevance Scores

	★	🚿	🍴	
A	3	N	Y	0.1
B	5	Y	Y	0.4
C	5	Y	N	0.4
D	3	N	N	0.1

- The formulation of querying **an attribute** can be written as:

$$\Psi_{\text{CG}}(\text{query}(a)) = \sum_{w_a \in \mathcal{W}_a} \left(\Psi_{\text{CG}}(w_a = w_a^*) \cdot \Pr(w_a = w_a^*) \right),$$

where $w_a = w_a^*$ means that when querying a , the user's answer (represented by w_a^* is w_a).

Online Checking Module

- The resulting online decision tree algorithm is to greedily choose an attribute (value) or an item that can maximize the certainty gain:

$$a_{\text{query}} = \arg \max_{a \in \mathcal{V}_{k-1} \cup \mathcal{X}_{k-1}} \Psi_{\text{CG}}(\text{query}(a)),$$

(a) Matrix of Candidate Items with Offline Estimated Relevance Scores

	★	🚿	🍴	
A ★★★	3	N	Y	0.1
B ★★★★	5	Y	Y	0.4
C ★★★★★	5	Y	N	0.4
D ★★★	3	N	N	0.1

- The formulation of querying **an item** is

$$\begin{aligned} \Psi_{\text{CG}}(\text{query}(a)) &= \Psi_{\text{CG}}(a \in \mathcal{V}^*) \cdot \Pr(a \in \mathcal{V}^*) + \Psi_{\text{CG}}(a \notin \mathcal{V}^*) \cdot \Pr(a \notin \mathcal{V}^*) \\ &= \left(\sum_{v_m \in \mathcal{V}_{k-1}} \Psi_{\text{RE}}(v_m) \right) \cdot \Pr(a \in \mathcal{V}^*) + \Psi_{\text{RE}}(a) \cdot \Pr(a \notin \mathcal{V}^*), \end{aligned}$$

where $a \in \mathcal{V}^*$ means that queried a is **a target item**.

Online Checking Module

We reveal the connection between querying **attributes** (i.e., querying **attribute IDs**) and querying attribute values in the following proposition.

Proposition 1. *For any attribute $x \in \mathcal{X}_{k-1}$, $\Psi_{\text{CG}}(\text{query}(x)) \geq \Psi_{\text{CG}}(\text{query}(x) = w_x)$ holds for all the possible $w_x \in \mathcal{W}_x$.*

In the context of querying attribute values, we further reveal what kind of attribute value is an ideal one in the following proposition.

Proposition 2. *In the context of querying attribute values, an ideal choice is always the one that can partition all the unchecked relevance scores into two equal parts (i.e., the ideal $w_x \in \mathcal{W}_x, x \in \mathcal{X}_{k-1}$ is the one that makes $\Psi_{\text{CG}}(w_x = w_x^*) = \text{SUM}(\{\Psi_{\text{RE}}(v_m) | v_m \in \mathcal{V}_{k-1}\})/2$ hold), if it is achievable. And the certainty gain in this case is $\Psi_{\text{CG}}(\text{query}(x) = w_x) = \text{SUM}(\{\Psi_{\text{RE}}(v_m) | v_m \in \mathcal{V}_{k-1}\})/2$.*

To get rid of the impact of the recommender system, we introduce a cold-start setting, where $\Psi_{\text{RE}}(v_m) = 1/M$ holds for any $v_m \in \mathcal{V}$.

Lemma 1. *In the context of querying attribute values, suppose that $\Psi_{\text{RE}}(v_m) = 1/M$ holds for any $v_m \in \mathcal{V}$, then the expected number of turns (denoted as $\hat{K}$) is bounded by $\log_2^{M+1} \leq \hat{K} \leq (M+1)/2$.*

Convolutional recommender systems (purely recommending an item at each turn) can be regarded as a **special case** of our framework.

Online Checking Module

- We notice that the above formulations of either querying attributes or querying attribute values, **does not consider the dependence among attributes** (e.g., as Figure 1(a) shows, attribute **Hotel Level** can largely determine attribute **Shower Service**).

$$\Psi_{\text{CG}}^{\text{D}}(\text{query}(a)) = \sum_{a' \in \mathcal{X}_{k-1}} \left(\Psi_{\text{CG}}(\text{query}(a')) \cdot \Pr(\text{query}(a') | \text{query}(a)) \right),$$

$\Pr(\text{query}(a') | \text{query}(a))$ measures the probability of the user preference on a determining the user preference on a' .

$$\Psi_{\text{CG}}^{\text{D}}(\text{query}(a)) = \sum_{w_a \in \mathcal{W}_a} \left(\Psi_{\text{CG}}^{\text{D}}(w_a = w_a^*) \cdot \Pr(w_a = w_a^*) \right),$$

where we define $\Psi_{\text{CG}}^{\text{D}}(w_a = w_a^*)$ as:

$$\Psi_{\text{CG}}^{\text{D}}(w_a = w_a^*) = \sum_{a' \in \mathcal{A}} \sum_{w_{a'} \in \mathcal{W}_{a'}} \left(\Psi_{\text{CG}}(w_{a'} = w_{a'}^*) \cdot \Pr(w_{a'} = w_{a'}^* | w_a = w_a^*) \right),$$

where $\Pr(w_{a'} = w_{a'}^* | w_a = w_a^*)$ measures the probability of how likely getting the user preference on attribute a determinates the user's preference on other attributes a' .

Offline Training Module

- Our paper lists two separate configurations in terms of querying attributes: one is directly querying **attribute IDs** (e.g., querying a user what **level of hotel** she prefers), while the other one is querying **attribute values** (e.g., querying a user whether she prefers hotels with **level 3**).

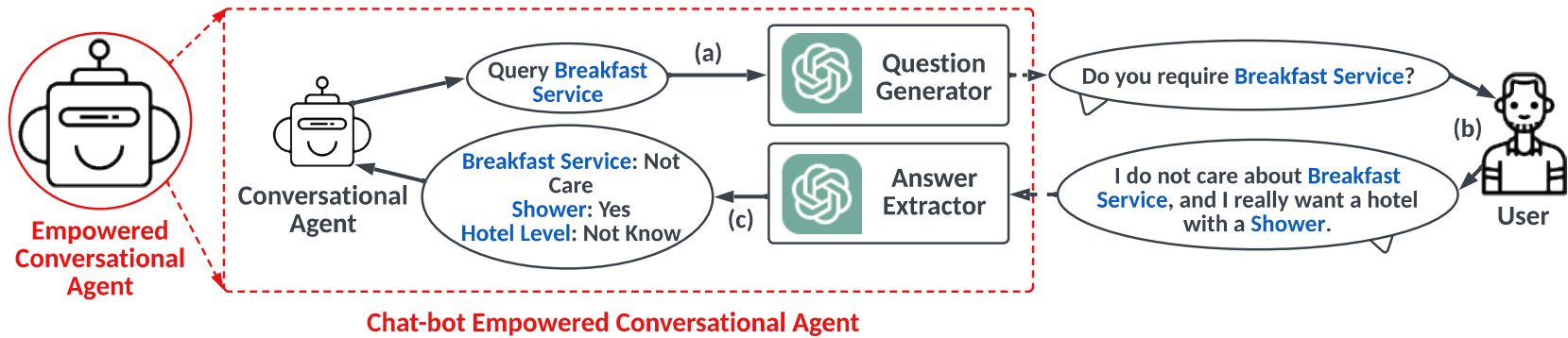
Algorithm 1 CORE for Querying Items and Attributes

Input: A recommender system $\Psi_{\text{RE}}(\cdot)$, an item set $\mathcal{V}$, an attribute set $\mathcal{X}$, an offline dataset $\mathcal{D}$.

Output: Updated recommender system $\Psi_{\text{RE}}(\cdot)$, up-to-date dataset $\mathcal{D}$.

- 1: Train $\Psi_{\text{RE}}(\cdot)$ on $\mathcal{D}$. ▷ Offline-Training
 - 2: **for** each session (i.e., the given user) **do**
 - 3: Initialize $k = 1$ and $\mathcal{V}_0 = \mathcal{V}$, $\mathcal{X}_0 = \mathcal{X}$.
 - 4: **repeat**
 - 5: Compute a_{query} following Eq. (3) for querying items and attributes or following Eq. (10) for querying items and attribute values. ▷ Online-Checking
 - 6: Query a_{query} to the user and receive the answer. ▷ Online-Checking
 - 7: Generate $\mathcal{V}_k$ and $\mathcal{X}_k$ from $\mathcal{V}_{k-1}$ and $\mathcal{X}_{k-1}$. ▷ Online-Checking
 - 8: Go to next turn: $k \leftarrow k + 1$.
 - 9: **until** Querying a target item or $k > K_{\text{MAX}}$ where K_{MAX} is the maximal number of turns.
 - 10: Collect session data and add to $\mathcal{D}$.
 - 11: **end for**
 - 12: Update $\Psi_{\text{RE}}(\cdot)$ using data in $\mathcal{D}$. ▷ Offline-Training
-

Prompt Design for Extractor and Generator



- Why use LLM here:
 - LLM is a powerful language model for understanding and generating conversations.
 - LLM can encode some **latent features** from conversations to guide the decision-making of the conversational agent.

Prompt Design for Extractor

```
"""
prompt = f"""
You will be provided with text delimited by triple quotes.
If you can infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: [user preference]
Attribute names include Breakfast Service, Hotel Level, and Shower.
User preference includes Yes to denote the positive preference, No to denote the
negative preference, and Not Care to denote the user does not care.
If you can not infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: Not Know
““{text}““
"""

response = get_completion(prompt)
print(response)
```

The following is the corresponding output by the LLM.

```
Breakfast Service: Not Care
Hotel Level: Not Know
Shower: Yes
```

Prompt Design for Extractor

```
# Large language model as latent feature extractor.
text = f"""
I am a student, I do not care about breakfast service, and I really want a hotel with a shower.
"""

prompt = f"""
You will be provided with text delimited by triple quotes.
If you can infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: [user preference]
Attribute names include Breakfast Service, Hotel Level, Price, and Shower.
User preference includes Yes to denote the positive preference, No to denote the
negative preference, and Not Care to denote the user does not care.
If you can infer the user has a low budget, please make Price: Low.
If you can not infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: Not Know
'''{text}'''
"""

response = get_completion(prompt)
print(response)
```

The following is the corresponding output by the LLM.

```
Breakfast Service: Not Care
Hotel Level: Not Know
Price: Low
Shower: Yes
```

Prompt Design for Generator

```
# Large language model as question generator.
text = f"""
Attribute, Hotel Level is 5, Hotel
"""

prompt = f"""
You will be provided with text delimited by triple quotes.
If it starts with the word "item", it denotes an item \
then you should generate a text to recommend the item to the user.
Otherwise, it denotes an attribute \
then you should generate a text to query the user's preference on the attribute.
You should be gentle.
''{text}''
"""

response = get_completion(prompt)
print(response)
```

The following is the corresponding output by the LLM.

Excuse me, may I ask for your preference on hotel level? Would you prefer a 5-star hotel or are you open to other options?

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Experiment

Table A1: Results comparison in the context of querying attributes and items on tabular datasets.

$\Psi_{RE}(\cdot)$	$\Psi_{CO}(\cdot)$	Amazon				LastFM				Yelp			
		T@3	S@3	T@5	S@5	T@3	S@3	T@5	S@5	T@3	S@3	T@5	S@5
COLD START	AG	6.47	0.12	7.83	0.23	6.77	0.05	8.32	0.14	6.65	0.08	8.29	0.13
	ME	3.04	0.98	5.00	1.00	3.00	1.00	5.00	1.00	3.00	1.00	5.00	1.00
	CORE	2.88	1.00	2.87	1.00	2.73	1.00	2.75	1.00	2.92	1.00	2.94	1.00
	CORE _D ⁺	2.84	1.00	2.86	1.00	2.74	1.00	2.73	1.00	2.90	1.00	2.91	1.00
FM	AG	2.76	0.74	2.97	0.83	4.14	0.52	4.67	0.64	3.29	0.70	3.39	0.81
	CRM	3.07	0.98	3.37	0.83	2.98	0.99	3.43	1.00	3.08	0.98	3.12	0.96
	EAR	2.98	0.99	3.13	1.00	3.02	1.00	3.51	1.00	2.94	1.00	3.02	0.99
	CORE	2.17	1.00	2.16	1.00	2.06	1.00	2.07	1.00	2.09	1.00	2.10	1.00
	CORE _D ⁺	2.14	1.00	2.14	1.00	2.05	1.00	2.05	1.00	2.10	1.00	2.08	1.00

[ME] always generates a query in the attribute level with the maximum entropy principle.

[AG] always queries an item with the highest relevance score at each turn, which is equivalent to solely using recommender systems.

[CRM] [Yueming Sun, et al.] integrates the conversational component and the recommender component by feeding the belief tracker results to the recommendation method.

[EAR] [Wenqiang Lei, et al.] consists of three stages, i.e., the estimation stage, the action stage, and the reflection stage.

[FM] [Steffen Rendle] considers the second-order interactions among attributes (i.e., feature fields).

[DEEP FM] [Huifeng Guo, et al.] combines an FM component and a neural network component together to produce the final prediction.

[PNN] [Yanru Qu, et al.] includes an embedding layer to learn a representation of the categorical data and a representation of user interests from historical behaviors.

[CORE] is our method but not consider the dependence among attributes, and CORE_D⁺ considers.

Ref: Wenqiang Lei, et al. Estimation–Action–Reflection: Towards Deep Interaction Between Conversational and Recommender Systems. WSDM, 2020.

Ref: Yueming Sun, et al. Conversational Recommender System. SIGIR, 2018.

Ref: Steffen Rendle. Factorization Machines. ICDM, 2010.

Ref: Huifeng Guo, et al. DeepFM: A Factorization-machine based Neural Network for CTR Prediction. IJCAI, 2017.

Ref: Yanru Qu, et al. Product-based Neural Networks for User Response Prediction. ICDM, 2016.

Experiment

Table 2: Results comparison of querying attribute values on tabular datasets. See Table A2 for the full version.

$\Psi_{RE}(\cdot)$	$\Psi_{CO}(\cdot)$	Amazon				LastFM				Yelp			
		T@3	S@3	T@5	S@5	T@3	S@3	T@5	S@5	T@3	S@3	T@5	S@5
COLD START	AG	6.47	0.12	7.83	0.23	6.77	0.05	8.32	0.14	6.65	0.08	8.29	0.13
	ME	6.50	0.12	8.34	0.16	6.84	0.04	8.56	0.11	6.40	0.15	8.18	0.20
	CORE	6.02	0.25	6.18	0.65	5.84	0.29	5.72	0.74	5.25	0.19	6.23	0.65
	CORE _D ⁺	6.00	0.26	6.01	0.67	5.79	0.30	5.70	0.75	5.02	0.21	6.12	0.68
FM	AG	2.76	0.74	2.97	0.83	4.14	0.52	4.67	0.64	3.29	0.70	3.39	0.81
	CRM	4.58	0.28	6.42	0.38	4.23	0.34	5.87	0.63	4.12	0.25	6.01	0.69
	EAR	4.13	0.32	6.32	0.42	4.02	0.38	5.45	0.67	4.10	0.28	5.95	0.72
	CORE	3.26	0.83	3.19	0.99	3.79	0.72	3.50	0.99	3.14	0.84	3.20	0.99
DEEP FM	CORE _D ⁺	3.16	0.85	3.22	1.00	3.75	0.74	3.53	1.00	3.10	0.85	3.23	1.00
	AG	3.07	0.71	3.27	0.82	3.50	0.68	3.84	0.79	3.09	0.74	3.11	0.88
	CRM	4.51	0.29	6.32	0.40	4.18	0.38	5.88	0.63	4.11	0.23	6.02	0.71
	EAR	4.47	0.30	6.35	0.43	4.01	0.37	5.43	0.69	4.01	0.32	5.74	0.75
PNN	CORE	3.23	0.85	3.22	0.99	3.47	0.81	3.34	1.00	2.98	0.93	3.11	1.00
	AG	3.02	0.74	3.10	0.87	3.44	0.67	3.53	0.84	2.83	0.77	2.82	0.91
	CORE	3.01	0.88	3.04	0.99	3.10	0.87	3.20	0.99	2.75	0.88	2.76	1.00

Ref: Wenqiang Lei, et al. Estimation–Action–Reflection: Towards Deep Interaction Between Conversational and Recommender Systems. WSDM, 2020.

Ref: Yueming Sun, et al. Conversational Recommender System. SIGIR, 2018.

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Ref: Yanru Qu, et al. Product-based Neural Networks for User Response Prediction. ICDM, 2016.

Open-Source System

Definition A1 (Conversational Agent and Human User Interactions). *Our conversational agent is designed to act in the following four ways.*

- (i) *Query an item $v \in \mathcal{V}_{k-1}$, where $\mathcal{V}_{k-1}$ is the set of unchecked items after $k - 1$ interactions (e.g., recommend Hotel A to the user).*
- (ii) *Query an attribute $x \in \mathcal{X}_{k-1}$, where $\mathcal{X}_{k-1}$ is the set of unchecked attributes after $k - 1$ interactions (e.g., query what Hotel Level does the user want).*
- (iii) *Query whether the user's preference on an attribute $x \in \mathcal{X}_{k-1}$ is equal to a specific attribute value $w_x \in \mathcal{W}_x$ where $\mathcal{W}_x$ is the set of values of attribute x (e.g., query whether the user likes a hotel with Hotel Level=5).*
- (iv) *Query whether the user's preference on an attribute is not smaller than a specific value $w_x \in \mathbb{R}$ (e.g., query whether the user likes a hotel with Hotel Level $\geq$ 3.5).*

The human user is supposed to respond in the following ways.

- (i) *For queried item v , the user should answer Yes (if v satisfies the user) or No (otherwise) (e.g., answer Yes, if the user likes Hotel A).*
- (ii) *For queried attribute x , the user should answer her preferred attribute value, denoted as $w_x^* \in \mathcal{W}_x$ (e.g., answer 3 for queried attribute Hotel Level), or answer Not Care to represent that any attribute value works.*
- (iii) *For queried attribute value w_x , the user should answer Yes (if w_x matches the user preference) or No (otherwise) (e.g., answer Yes, if the user wants a hotel with Hotel Level=5), or answer Not Care to represent that any attribute value works.*
- (iv) *For queried attribute value w_x , the user should answer Yes (if the user wants an item whose value of attribute x is not smaller than w_x) or No (otherwise) (e.g., answer Yes, if the user wants a hotel with Hotel Level=5), or answer Not Care to represent that any attribute value works.*

Open-Source System

- Now we support **top-N** recommendations for items and attribute values.

CORE Public

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main 1 branch 0 tags Go to file Add file Code

Jinjiarui Update index.rst 86d8eb3 on Jul 27 57 commits

File	Commit Message	Time
assets	Add files via upload	3 months ago
docs	Update index.rst	3 months ago
system	minor	3 months ago
.gitignore	debug	3 months ago
LICENSE	debug	3 months ago
README.md	Update README.md	3 months ago

README.md

Conversational Agent for Recommender System (CORE)

A Plug-and-Play Conversational Agent for Recommender System

[Project](#) | [Paper](#) | [Documentation](#)

CORE is a plug-and-play conversational agent for any recommender system built upon [PyTorch](#). CORE is:

- Comprehensive:** CORE provides a data manager, an offline trainer, and an online checker.
- Flexible:** CORE could be integrated into any recommender system.
- Efficient:** CORE is built upon an online decision tree algorithm instead of heavily learning algorithms.

About

CORE is a plug-and-play conversational agent for any recommender system.

jinjiarui.github.io/core/

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Conclusion

- CORE only requires **estimated scores** from recommender systems (offline-training) and **APIs** from conversational agent (online checking).
- LLMs can plug-in specific task-related **tools** to encode **user's historical data and tabular data**.

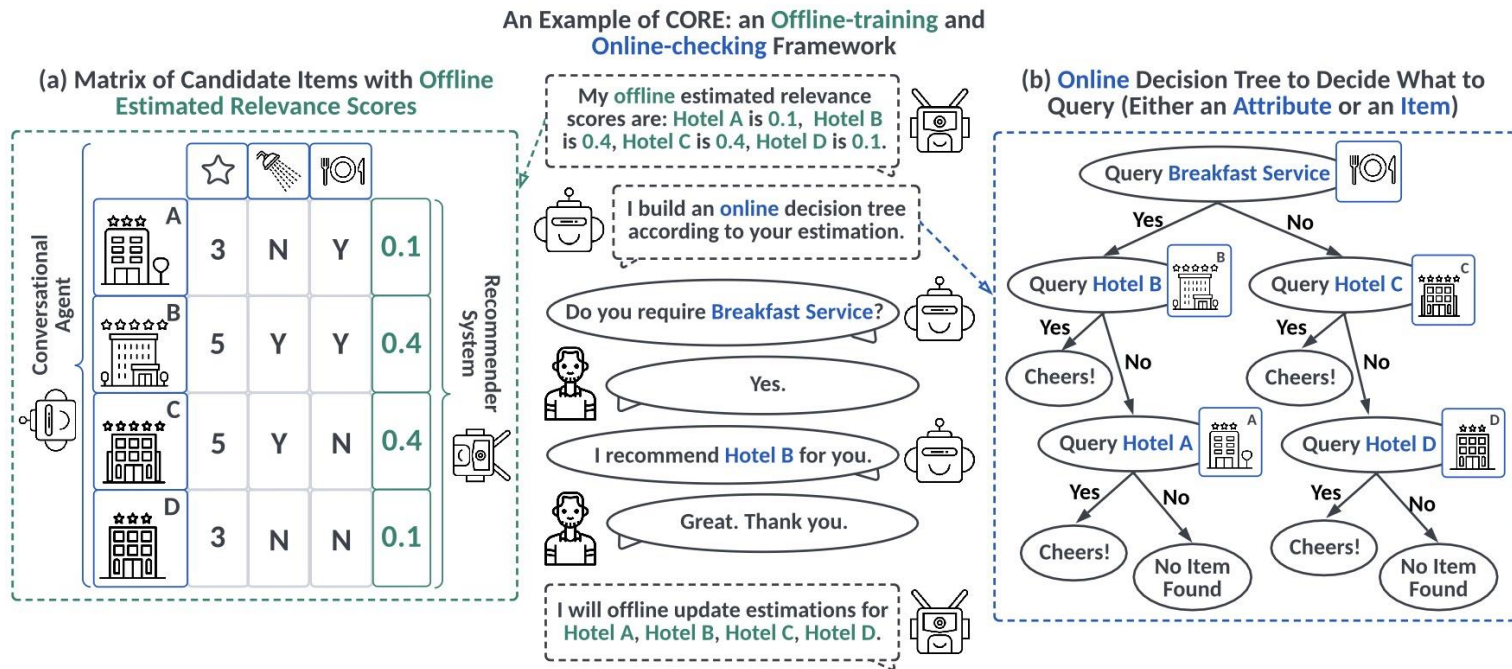


Figure: Overview of our offline-training and online-checking framework.

Thanks for Your Listening



Paper Link



Project Link



Slides Link



Poster Link